

AAKASH GUPTA

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SUMMARY

Five years building production software at **1.4M MAU** scale, now building applied AI: retrieval with citations, code review agents, and real-time voice. I build the system around the model, measure latency per stage, and keep facts out of the weights.

AI & GENAI PROJECTS

Codewalk ↗: AI Code Intelligence Platform · [demo video](#)

Indexing, RAG-based chat, and multi-agent code review over a repository, as one product

- **Multi-agent review:** a diff is partitioned into batches and a pool of review agents works them in parallel (**4 concurrent**), each batch returning findings plus its own coverage verdict, so a review with a failed batch reports as incomplete instead of passing. Findings are grounded against the checked-out revision before storage, then written back as PR comments and a check run.
- **RAG-based chat:** a question expands into 2 to 4 angles, then one shared gather pulls from a graph symbol walk and a batched vector search in a single embedding request; every candidate is scored **0 to 10** in one ranking call, which is the only step that drops evidence, and a failed ranking keeps everything. Answers cite the file and symbol.
- **Indexing:** tree-sitter parses **14** languages into a symbol graph and parent-child chunks, embedded under an adaptive rate governor, then promoted by an atomic swap so readers only ever open a finished index and a retry resumes rather than re-embedding files it already paid for.

Stack: Python, FastAPI, PostgreSQL, Redis, tree-sitter, ChromaDB, DuckDB, Next.js, Docker, Go edge gateway

VoiceFlow ↗: Real-Time Voice AI Platform · [demo video](#)

- **Cascaded streaming pipeline** placing real phone calls: Twilio Media Streams → Deepgram STT → Grok/Kimi (function calling) → Cartesia/ElevenLabs TTS over WebSockets, 8 kHz mu-law end to end
- **Barge-in and turn-taking:** STT endpointing for turn detection; cancellable speak tasks and Twilio clear events so a caller can interrupt mid-sentence
- **Latency engineering:** sentence-level LLM→TTS streaming for low time-to-first-audio; per-turn STT/LLM/TTS/E2E average and p95 on a live dashboard, tagged with provider and model
- **Workflows and captures:** loan recovery, EMI, banking and sales as configuration; tools persist structured promise-to-pay, lead-qualify and escalate results; EN/Hi/Hinglish with language-aware STT and TTS voices

Stack: Python, FastAPI, WebSockets, Twilio, Deepgram, Grok (xAI), Cartesia, ElevenLabs, SQLite, Next.js, Render · Vercel

PRODUCTION EXPERIENCE

ANKO GCC · Mobile Engineer

Oct 2024 to Present

*Kmart & Target Australia, retail app at **3.74M** active users*

- **Performance engineering:** WebView memory **2GB → 400MB**; token clear **2,000ms → 10ms**; PLP first load **6 to 9s → 131ms**; PDP logged-in **3,054ms → 231ms**
- **Next-Gen Home Screen:** co-designed Contentful-driven architecture, shared BLoC patterns across **10+** modules: **38.9M** sessions, no major defects post-release
- **Shoppable UGC (Tolstoy):** owned SDK integration and vendor rollout: **17M** events, **923K** users, **24%** conversion among interactors; shipped 1 month early
- **Connectivity and standards:** screen-aware offline handling with auto-recovery, **2 to 7s** faster reconnect; ran BLoC and code-quality brown-bags for **10 to 12** engineers

Explorex Technologies · Frontend Developer (Flutter)

Oct 2023 to Jul 2024

- Built Digital Dining (home, menu, cart, pay-bill) from scratch, live at **300+ restaurants** in Bangalore

Deciml · Flutter Developer

May 2023 to Aug 2023

- Migrated FAQs/blogs to Firebase Remote Config; owned bug fixes, releases, and product-copy updates

Threedots · Product Engineer

Sep 2021 to Mar 2023

Social investing platform, 100K+ users

- Built Trade Feeds (**50K users**), Tag-Based Group Discovery (**24K**), In-App Rating (**300K**), current-affairs feed (**34K**)
- Demo-gameplay onboarding: **63.5K** interactions, **+10% new-user engagement**; refactored Groups to BLoC, cutting load time **50%**

SKILLS

AI / GenAI: RAG, retrieval and ranking, embeddings, chunking, agent runtimes, function calling, prompt and context design, streaming voice pipelines, latency instrumentation, evaluation harnesses, LangChain, LangGraph, ChromaDB, DuckDB, Langfuse

Backend & Real-time: Python, FastAPI, WebSockets, SSE, async services, job state machines, idempotency, retries with backoff, circuit breakers, PostgreSQL, Redis, SQLite, Docker, CI/CD

Mobile & Languages: Flutter (BLoC), SwiftUI, Firebase · Python, Dart, Swift, SQL

EDUCATION

Thakur College of Engineering, B.E. IT (2018 to 2021) · Government Polytechnic Mumbai, Diploma IT (2015 to 2018)